

Requirements Extraction - AI-Assisted Elicitation and Analysis

Source stakeholder statement

"We're a mid-sized clinic. Booking is a mess - patients call and we write appointments in a paper diary, so we double-book and forget reminders. We'd like something online, but most of our patients are elderly. And whatever we do has to keep their records private."

Simulated local-LLM first draft

Stated needs

- Replace or improve the paper-based appointment-booking process.
- Provide an online capability for appointments.
- Reduce double-booking.
- Improve appointment reminders.
- Protect patient records and privacy.
- Account for an elderly patient population when designing the service.

Implied needs

- A centralized appointment schedule so staff see the same current availability.
- Validation that prevents two appointments from occupying the same constrained slot/resource.
- A reminder mechanism, potentially using one or more communication channels.
- An accessible and simple user experience.
- Authentication and authorization appropriate to the information exposed.
- A staff workflow for creating, changing and cancelling appointments.

Open questions

1. Who books appointments today - reception staff, clinicians, patients, or all three?
2. Should patients book directly online, request an appointment for staff approval, or both?
3. What appointment types, durations, clinicians, rooms or resources affect availability?
4. What does "double-book" mean operationally, and are there any intentional exceptions?
5. Which reminder channels are acceptable: SMS, email, telephone, app notification, or another method?
6. How far in advance should reminders be sent, and can patients opt out?
7. Must telephone booking remain available for patients who cannot or do not want to use the online service?
8. What accessibility requirements should the web/mobile experience satisfy?

9. What patient information is required to book, and what information should never be displayed unnecessarily?
10. Which privacy, security, retention, audit and regulatory obligations apply in the clinic's jurisdiction?
11. Do existing patient-record, billing, calendar or messaging systems need to integrate with the solution?
12. What volumes, operating hours, locations and availability rules must the system support?
13. How will success be measured - fewer booking conflicts, fewer missed appointments, shorter call times, or another KPI?

Manual BA corrections and verification

The AI draft is useful for generating hypotheses, but several points require correction or qualification.

- **Correction - "online" does not automatically mean a mobile app.** The client only asked for something online. A responsive website, patient portal, or other channel remains an open solution decision.
- **Correction - reminders are a problem, but the channel is unknown.** SMS or email must not be converted into a requirement until stakeholders select an approved channel and consent/privacy implications are understood.
- **Added item the draft could miss - assisted/offline access must be investigated.** Because many patients are elderly, the BA should explicitly ask whether telephone or staff-assisted booking must remain available. Elderly age alone must not be treated as proof that patients cannot use technology.
- **Qualification - "prevent all double-booking" may be wrong.** Some clinics intentionally overbook or allow exceptions. The business rule needs to be elicited rather than assumed.
- **Qualification - privacy is stated, but the exact compliance framework is not.** The BA should not invent a particular law without confirming the clinic's location, data handled, organizational obligations, and approved security/privacy standards.

Curated requirement candidates

The following are candidate requirements, not stakeholder-approved requirements:

- The solution shall maintain a centralized view of appointment availability for authorized users.
- The solution shall detect booking conflicts according to stakeholder-approved scheduling rules.
- The solution shall support an online appointment workflow whose exact booking/approval model is to be confirmed.
- The solution shall support stakeholder-approved reminder workflows.
- The user experience shall address validated accessibility and usability needs.

- The solution shall restrict access to patient information according to approved roles and privacy/security requirements.
- The solution shall support required staff workflows for creating, modifying and cancelling appointments.
- Continued telephone or assisted booking shall be evaluated as a requirement during discovery.